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Asymmetric Relations Between Needs and Data Availability in Predictive Maintenance Systems

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Abstract

Maintenance decisions can remain consequential when feasible condition information is persistently weak. This paper characterizes Limited-Data Predictive Maintenance (LD-PdM) as a regime in which proxy, contextual, historical, and prior evidence materially informs the assessment of maintenance needs. It develops an applicability checklist, a five-step assessment procedure, and a railway-station inspection example. The checklist separates weak condition information from small sample count. The procedure connects evidence and assumptions to admissible actions, including a fallback when a comparison remains unresolved. A two-door model evaluates early inspection together with the response after a positive finding, relative to an existing schedule. Exact calculations with hypothetical inputs give regions where exposure ordering is preserved, reversed by detection differences, or unresolved across routing interpretations. They also expose a cost of requiring unanimous preference: preserving the schedule can forgo an inspection that improves on it under both models. A shared-error calculation separates aggregation uncertainty from pairwise ranking. The contribution organizes established small-data PHM, usage-based maintenance, soft sensing, and probabilistic decision analysis around a specified choice and its feasible evidence. The numerical results illustrate conditional decisions; they do not estimate station performance, calibrated failure probabilities, or industrial savings.

Keywords: predictive maintenance; limited data; partial observability; proxy information; Bayesian inference; uncertainty; risk-based maintenance; railway stations; inspection planning; decision support

1. Introduction

Predictive maintenance (PdM) connects evidence about an asset to decisions about its future operation, inspection, servicing, or replacement. Diagnosis and remaining useful life (RUL) prediction are important components of this process, but the operational purpose is to choose an appropriate action. Reviews describe the development of PdM, machine-learning methods, and their relationship to industrial information infrastructures [1–3]. The availability of informative measurements influences which parts of this workflow can be implemented. It does not, however, determine whether a maintenance decision has to be made.

Limited evidence is already a recognized problem in prognostics and health management (PHM). Li et al. [4] review limitations in monitoring quantity and quality, economic and operational barriers, and data-efficient learning. Their scope includes maintenance execution, multimodal records, physical knowledge, fleet information, and uncertainty.

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The question here is how to assess a maintenance proposal when feasible condition measurements leave important uncertainty and the proposed action depends on interpreting operational records, service history, or engineering assumptions.

This regime is termed *Limited-Data Predictive Maintenance* (LD-PdM). The term describes the evidence available for a maintenance choice. Consider infrastructure devices whose maintenance affects service continuity, while the feasible information budget provides little dedicated condition monitoring for each device. Passenger flows, event logs, service records, documentation, and engineering knowledge may nevertheless inform hypotheses about usage and maintenance needs. The central asymmetry is that the consequence of a maintenance choice and the availability of informative evidence need not increase together. High decision consequences do not ensure informative observations; abundant records do not ensure relevance to the decision.

Established decision analysis provides the basis for comparing the resulting actions. Srinivasan and Parlikad [5] assess condition-monitoring value for infrastructure maintenance using a partially observable Markov decision process (POMDP), while Memarzadeh and Pozzi [6] connect information value to inspection, monitoring, and system-level scheduling. Bayesian-network applications in safety, reliability, and risk assessment likewise provide established tools for integrating uncertain evidence [7]. The present assessment uses these foundations to check what the available records can support before committing to an inference method or maintenance policy.

The paper develops three practical outputs for examining such a maintenance proposal: an applicability checklist (Table 1), a five-step assessment procedure (Section 4.1), and a worked two-door inspection comparison (Section 5.3). The comparison derives and numerically illustrates when an exposure-based order agrees with a consequence-based choice, when it reverses, and when the evidence cannot support a common strict preference. These outputs organize established PHM, indirect-sensing, and decision-analysis methods around the available evidence and the operator's remaining choice. The contribution is a conceptual synthesis; the station and inspection assumptions are hypothetical.

Section 2 defines the regime and its asymmetries. Section 3 describes evidence and belief formation, and Section 4 connects them to maintenance choices. Section 5 works through a hypothetical station example. Section 6 discusses overlap with existing approaches, limitations, and evaluation questions; Section 7 concludes.

2. The LD-PdM Regime and Its Asymmetries

2.1. Operational definition and boundary cases

LD-PdM is defined relative to an asset or population, a maintenance choice, and a planning horizon. It applies when the following conditions hold together:

(1) The maintenance choice has a consequential operational or economic outcome.
(2) A condition quantity relevant to that choice remains weakly informed by the feasible set of condition-sensitive channels because of persistent deployment, access, or acquisition constraints.
(3) Proxy, contextual, historical, or prior evidence therefore makes a material contribution to the assessment of future maintenance needs.

Here, persistent means that the limitation is expected to remain relevant over the decision horizon. It need not be permanent. The same asset can move between regimes if its instrumentation, data access, operating context, or maintenance objective changes.

Table 1 turns the criteria into questions requiring case-specific evidence. For each row, record *yes*, *no*, or *unresolved*, with the justification and its source. All three affirmative answers establish applicability under the stated assessment. A negative answer excludes the choice from this definition; an unresolved answer leaves classification open. Neither

Table 1. Applicability checklist for a specified maintenance choice and horizon. Each row requires a yes/no/unresolved answer supported by the indicated justification.

Criterion	Assessment question	Required justification
Consequential choice	Could admissible actions differ in service, safety, resource use, or economic outcome?	Name the affected function, alternatives, horizon, and consequence that makes the distinction matter.
Persistently weak condition information	Do feasible condition-sensitive channels leave a relevant condition distinction unresolved over that horizon?	Identify the quantity, available channels, remaining uncertainty, and constraint preventing its resolution.
Material indirect evidence	Does proxy, context, history, or prior information affect the assessment of future maintenance needs?	Show which inference or justification changes when that evidence or assumption changes. Mere availability is insufficient.

outcome establishes whether a particular inspection order is justified. That requires the separate action comparison in Section 4.

For example, a door may have a reliable indication of a presently obstructed opening but no informative channel for wear affecting the next inspection interval. Clearing the observed obstruction and choosing which door to inspect early are different decisions about the same asset. The first need not satisfy the second criterion; the latter may do so. Similarly, few labelled samples with a sufficiently informative condition channel create a small-data learning problem without necessarily creating LD-PdM. Conversely, extensive traffic records may leave a wear mechanism weakly informed. Dedicated measurements also require interpretation, so the distinction concerns their relevance to the choice, rather than whether condition is inferred at all.

The four asymmetries have different logical roles. Observability and acquisition constraints explain the defining mismatch between maintenance need and feasible condition information. Sparse or selective failure records complicate learning within that regime. Population scale creates opportunities for sharing evidence and coordinating work, together with additional dependencies. Rare failures and large populations are therefore not additional inclusion requirements. Low net information value is also not a measure of weak observability: information content, its effect on a choice, and acquisition cost must be assessed separately.

2.2. *Asymmetry of observability*

The operational importance of a variable does not guarantee an informative measurement channel. A planner may need to anticipate loss of service while observing only traffic, location, occasional alarms, or service interventions. Such records can constrain hypotheses about device usage without identifying condition. Figure 1 separates the available evidence from the model-dependent quantities required for decisions.

In the formulation of Tinga [8], usage- and load-based maintenance connects operating information to condition and maintenance intervals through physical failure models. Soft sensors estimate process quantities from other measurements using model-driven or data-driven relationships, with model validation and changing operating conditions as practical concerns [9]. In the station setting, even device usage may be indirectly known. The assessment must therefore identify both the target of each estimate and the evidence supporting the next link to maintenance need.

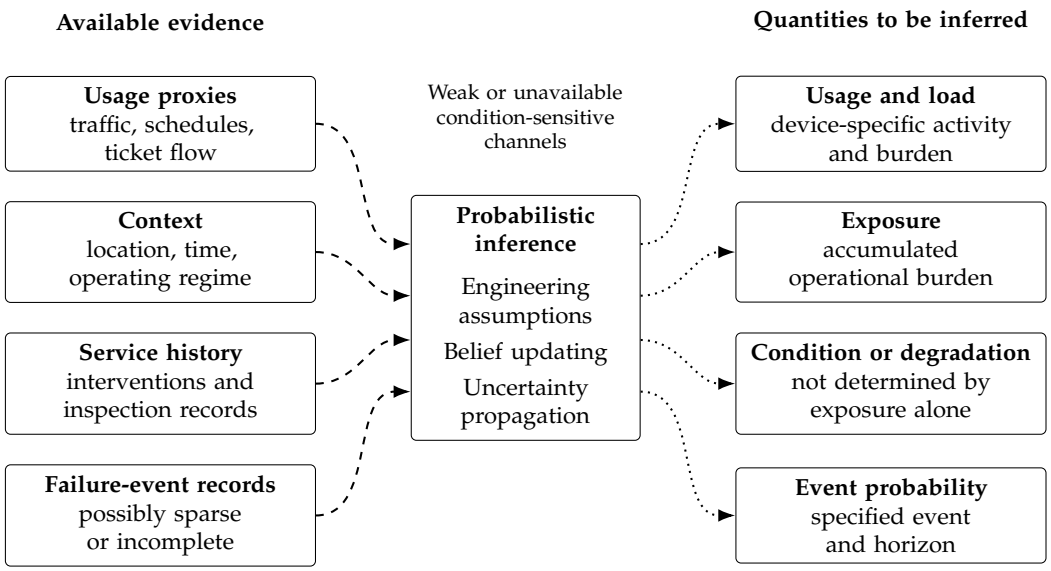


Figure 1. Conceptual separation between evidence, probabilistic reasoning, and maintenance quantities. Dashed arrows denote evidence inputs; dotted arrows denote inferential outputs. The common inference block leaves individual source-to-target dependencies unspecified. All links require model assumptions; informative usage proxies need not resolve condition or event probabilities.

2.3. Asymmetry of failure evidence

The evidence generated by operation and maintenance can differ from the evidence required to learn failure behaviour. Rare faults provide few examples, operating conditions differ, and intervention changes what is subsequently observed. A replacement before failure may leave a right-censored lifetime; periodic inspection may locate an event only within an interval. Missing work orders are a separate problem. A maintenance intervention also changes the physical state, so it cannot always be represented as an observation ending without consequence.

Statistical work on interval-censored data provides machinery for partially observed event times [10], but its applicability depends on the observation assumptions. In maintenance, inspections and replacement may be triggered by suspected deterioration, making the observation process informative. Accordingly, absence of a recorded failure should not automatically be interpreted as evidence of healthy condition. A conceptual framework must preserve the distinction between survival, intervention, and missing records; a later quantitative model must justify its treatment of each.

2.4. Asymmetry of information costs and decision consequences

The value of a maintenance decision and the cost of improving its evidence are different quantities. The latter includes acquisition and installation, integration with existing systems, commissioning and validation, connectivity and data processing, and the continuing operation and maintenance of the monitoring arrangement. The importance of deployment cost is also recognized in small-data PHM [4]. It motivates a conditional applicability statement: LD-PdM is relevant where better dedicated information cannot presently be obtained within the operator's feasible resources and constraints.

This condition must be assessed for the particular asset function and decision. Unit purchase price does not determine criticality, and an elevator or door may have substantial safety, accessibility, or service consequences. Additional monitoring can therefore be

Table 2. Quantities distinguished in the conceptual framework. The final column identifies an inference limit, rather than a missing numerical parameter to be supplied in this paper.

Quantity	Meaning	Interpretive limit
Evidence	Proxy observations, context, service records, documentation	Record volume does not establish condition information
Usage	Activity during a specified interval, such as opening cycles	Passenger traffic is not a device-cycle counter
Exposure	Accumulated mechanism-relevant burden over a stated history	Equal cumulative activity need not imply equal wear
Condition	Physical degradation or functional state	Similar exposure does not imply equal condition
Event probability	Probability of a specified event within a declared horizon	A service threshold and a physical failure are different events
Risk	Consequences evaluated together with event uncertainty	An event probability alone does not specify consequences
Priority and action	A choice given beliefs, consequences, and constraints	Priority is a decision output, not a physical state

justified for a consequential function even in a population of ordinary devices. The station example supplies no asset-class estimate of sensing profitability. Condition-monitoring value depends on what the information changes in the maintenance policy [5]; existing operational data also have integration and interpretation costs.

2.5. Asymmetry of scale

Uncertain individual states can coexist with useful population-level questions: where exposure concentrates, which groups merit inspection, or how limited attention should be distributed. Hierarchical modelling can share information across appropriately related units; Bayesian IoT failure modelling provides one example [11]. Risk-based and opportunistic maintenance studies illustrate the importance of system consequences and coordinated interventions [12,13]. These are established modelling opportunities, not a guarantee that a larger population makes weak evidence sufficient.

Partial pooling requires a defensible relationship between units, including their mechanisms, environments, and maintenance histories. A hierarchical model should allow between-unit variation where these differ; complete pooling would impose a stronger common-parameter assumption. Shared routing errors or a common misspecified proxy relationship do not disappear when more assets are included. Heterogeneous asset classes may require different models. Furthermore, estimating total burden and selecting the particular asset requiring action are different tasks: identical evidence and priors do not justify a data-derived individual ordering. The operator must therefore distinguish estimating population burden from choosing an individual asset for inspection.

3. Evidence and Belief Formation

3.1. Separating observations from maintenance quantities

A proxy is informative only through a relationship with a specified target. Passenger counts might constrain use of an access route; a device-use model might translate that flow into operating cycles; cycles might inform cumulative exposure. None of these links, by itself, measures damage or establishes a failure probability. Age, maintenance quality, environment, design, and latent defects may alter the relationship between exposure and condition. Table 2 distinguishes the objects needed to describe that chain.

The evidence may combine operational aggregates, contextual variables, inspection and intervention histories, and engineering documentation. Their roles should remain identifiable. A timetable constrains possible demand; a service record describes an inter-

vention; documentation supplies a mechanism or service requirement. Treating these as interchangeable features obscures the observation assumptions. Context-dependent heating, ventilation, and air-conditioning (HVAC) diagnostics illustrate the use of operating context [14], while maintenance decision systems illustrate heterogeneous information integration [15]. Their data requirements must be checked against the station's available condition information.

3.2. Representing the unresolved quantities

The assessment should identify which quantities are observed, inferred, or supplied by engineering judgment. For a door, usage can be the number of opening cycles in a recording interval, whereas exposure accumulates relevant activity over a stated history. A wear model may weight cycles differently according to load or environment. The history over which exposure is accumulated must be distinguished from the future calendar-time window over which the maintenance choice is evaluated. No universal conversion from passenger totals to either quantity is assumed here.

A probabilistic representation can connect these quantities while preserving uncertainty about their relationships. Bayesian updating supplies established tools for learning from observations [16]; Bayesian networks and expert systems can represent dependencies and judgment-based assumptions [7,17]. The representation should be as specific as the decision requires. Section 5.3 uses a probability of an actionable condition at inspection, together with detection and consequence assumptions. It does not require a full degradation trajectory or a physical-failure probability. Either extension would need an additional transition model and an explicit event definition.

3.3. Priors, dependence, and limits of identification

Engineering knowledge may constrain plausible routing, device-use mechanisms, deterioration sensitivities, or service conditions. Such information belongs in an explicit assumption register with its source and scope. Documentation, expert judgment, and previously observed behaviour need not have equal evidential strength. Multiple records derived from the same original traffic estimate are also not independent confirmations. A model should preserve their dependence rather than treating duplication as increased support.

When observations weakly constrain a parameter, its posterior can remain strongly dependent on the prior or chosen model class. Honest uncertainty representation therefore includes sensitivity to plausible alternative assumptions and identification of links that the available evidence cannot resolve. In LD-PdM, this can mean stopping at an exposure comparison rather than extending the chain to a numerical failure probability. Inspection or a targeted measurement may then be more useful than a more elaborate inference algorithm applied to the unchanged evidence.

3.4. Shared error and population comparisons

A simple error model separates aggregation from ranking. Suppose an exposure estimate has the form $\widehat{x}_i = x_i + B + \varepsilon_i$, where the true exposures x_i are treated as fixed, B is a shared zero-mean error with variance σ_B^2, and the asset-specific errors ε_i have zero mean and common variance σ_ε^2. Assume the asset-specific errors are mutually independent and independent of B. For n assets,

$$\mathrm{Var}(\overline{\widehat{x}} - \overline{x}) = \sigma_B^2 + \frac{\sigma_\varepsilon^2}{n}, \qquad \widehat{x}_i - \widehat{x}_j = (x_i - x_j) + (\varepsilon_i - \varepsilon_j). \tag{1}$$

The shared component remains in the population mean but cancels in a pairwise difference. A fixed additive bias instead remains in the mean error; it should not be described as a variance component. Group-specific routing errors need not cancel between groups. Thus a uniform offset can preserve an ordering while distorting total exposure, whereas an erroneous allocation between routes can distort the ordering itself. These elementary identities explain why more records from the same source need not resolve the uncertainty relevant to a station decision.

4. Maintenance Choices under Weak Information

4.1. A procedure for assessing a maintenance proposal

The procedure takes an asset function, a proposed maintenance choice, the available records, and the operator's constraints as inputs. Its output is a documented action rationale or an unresolved comparison with a specified fallback. It can be applied before selecting an inference algorithm:

1. **Specify the decision.** State the mechanism, future planning window, consequences, available actions, resource limit, and mandatory requirements. Record the existing admissible schedule as the reference choice.
2. **Assess applicability.** Complete Table 1. Identify feasible condition-sensitive channels and the quantity they leave unresolved. Explain why the constraint persists over the window and how indirect evidence affects the assessment.
3. **Trace the evidence.** Link each record to the quantity it informs. Record assumptions needed for the next inference, shared data sources, competing interpretations, and observations that could challenge them.
4. **Compare admissible choices.** Evaluate consequences under the stated plausible assumptions and equal resource constraints. An inspection choice includes the subsequent action after each possible finding. Check separately whether a policy improves on the reference schedule and whether it is preferred to the other alternatives throughout the model set.
5. **Record and act.** State the supported choice, its assumptions, and the trigger for reconsideration. If no common preference is supported, apply the documented admissible fallback. Request further information only when it can usefully arrive before the decision deadline. A new mandatory or consequence-based requirement takes precedence.

Classification, improvement over a baseline, and selection of a preferred action are separate judgments. A choice may satisfy the LD-PdM criteria while the available evidence supports no unique inspection order. Some policies may nevertheless improve on the reference schedule throughout the model set. The example adopts retention of that schedule as its fallback; Section 5.4 shows the gain this convention can forgo. An operational application must specify its conflict-resolution rule and record its consequences.

4.2. Comparing inspection policies

An inspection policy specifies both the initial inspection and the action after each possible observation. Let π denote such a policy, Π the admissible set, E the evidence at the planning review, and m an assumed model. Over the common future window, write

$$R_m(\pi) = \mathbb{E}_m^{\pi}[L \mid E], \qquad \pi_m^{\star} \in \arg\min_{\pi \in \Pi} R_m(\pi). \tag{2}$$

Here L is total loss over the declared window, including inspection, intervention, disruption, and any terminal consequence included in the comparison. The superscript indexes the prospective system evolution and observations under the chosen policy. It does not condition on an action selected by a historical policy. This distinction matters when past in-

spections were triggered by suspected deterioration. The finite policies in Section 5.3 make Equation 2 concrete without requiring a complete sequential optimizer. POMDP-based maintenance and information-value analyses provide the established decision-theoretic foundations [5,6].

An inspection has value when a possible finding changes a feasible subsequent action enough to justify the inspection burden. Greater uncertainty alone establishes neither this value nor a reason to intervene. A highly uncertain quantity may be irrelevant to the available actions, while a relatively well-understood defect can warrant prompt service. A counting exercise and a physical inspection also answer different questions: the former may resolve route allocation, while the latter may reveal a condition requiring action. Valuing either requires its observation model, cost, and possible effect on the subsequent decision.

4.3. Resources and robustness

Across assets, comparison requires a common objective and resource constraint. Exposure describes burden; expected loss also depends on the affected function; inspection value additionally depends on detection and what can be done after a finding. Section 5.3 shows how these distinctions can reverse an order even for two doors. Coordinated maintenance adds access and service dependencies [12,13,18,19]. A ranking of separate asset scores is insufficient where taking one asset out of service changes another's use or removes an accessible route.

A preference should be checked across a declared set of plausible models, rather than only under a single convenient routing interpretation. Agreement within that set supports a conditional recommendation; it does not establish that the set covers the true operating situation. If plausible alternatives disagree, the output should identify the disputed assumption and the fallback required by step 5. Stability, calibration, and operational benefit remain different properties.

5. Worked Conceptual Example: A Railway Station

5.1. Setting and assumed evidence

Consider a hypothetical station with a concourse, platform access routes, automatic doors, elevators, and escalators. The example assumes that dedicated condition-sensitive channels are insufficient for the maintenance choice under consideration. It does not assume that instrumentation is absent from every device or economically unjustified for these asset classes. Consequences depend on function and alternatives: an elevator serving an otherwise inaccessible route can require different treatment from one with equivalent accessible alternatives. Operational importance and safety requirements must therefore be assessed for each function, rather than assigned to an entire device class.

Suppose the planner has zone-level passenger counts or ticket-flow summaries, timetables and platform assignments, a station layout, device documentation, and dated inspection and service records. At the present review, there is capacity for at most one additional early inspection before the next scheduled review. A predefined service response after a positive finding is also assumed feasible. Existing mandatory work and the established service schedule remain in force. The future window is a common interval of calendar time; historical exposure is accumulated over a common past interval ending at the present review.

Applying Table 1 to this hypothetical inspection choice gives three conditional affirmative answers. Earlier discovery and treatment can affect service disruption or maintenance effort; feasible channels leave the targeted condition unresolved during the window; and routing and service-history assumptions affect the assessment of where an actionable condition may be found. Each answer depends on a stated premise of the example. If a reliable

condition indicator instead already determines the appropriate service response, the second answer changes to no for that decision. If the relevance of the passenger proxy cannot be justified even as a conditional model, the third answer is unresolved. The asset class alone determines none of these answers.

5.2. From passenger context to conditional beliefs

Passenger information first informs possible device use through an assumed routing relationship. Counts at an entrance do not identify the route taken by every passenger or the device selected. Timetables and platform assignments can constrain plausible routes, while closures and alternative access paths can change them. Routing uncertainty must remain attached to any resulting device-level usage belief. Ticket-flow and passenger-count summaries may also share an upstream source; their agreement is then weaker corroboration than agreement between independent observations.

Station topology gives this interpretation a system-level structure. A graph of entrances, circulation zones, platform connections, and accessible alternatives can encode which devices could serve a passenger stream without claiming to reconstruct individual trajectories. An outage or planned intervention may redirect that stream and change use of neighbouring assets. Device exposure is consequently conditional on route availability as well as demand. If the planner considers taking a device out of service, the resulting redistribution belongs to the consequences of that choice; the pre-intervention exposure map cannot simply be assumed to remain valid.

The next link is device-specific. For an automatic door, passage demand can inform an opening-cycle estimate only under assumptions about grouped passage, reopening, and control behaviour. For an elevator, passenger demand alone does not determine trips, occupancy, travel distance, or the distribution of loads. For an escalator, a continuously running operating regime differs from a demand-controlled regime: passenger flow and run time describe different aspects of use. Consequently, a common passenger indicator cannot be converted into a common wear score across these mechanisms.

Temporal structure matters for the same reason. Equal passenger totals spread evenly through the day or concentrated around train arrivals need not imply equal device activity. Where one opening serves several passengers, clustering can change the relation between passages and cycles. Where use is continuous, running duration may remain similar despite a change in passenger count. The relevant aggregation interval therefore follows the hypothesized mechanism and decision: a daily total can support a broad exposure discussion while concealing the timing needed to plan inspection without disrupting a concentrated demand period.

Usage may then inform a model of accumulated exposure over a stated interval, conditional on the mechanism being considered. Exposure is an intermediate description of operational burden, not an observation of damage. Connecting it to physical condition requires further assumptions about initial state, maintenance effects, environment, and the applicable degradation mechanism. A service record can modify those beliefs without establishing that the device was restored to an as-new state. Absence of a recorded fault is also compatible with incomplete reporting or earlier preventive intervention. Table 3 identifies where additional evidence could challenge the chain.

An event probability requires an additional, explicit event model. Reaching a documented service condition, discovering a defect at inspection, and losing required function before the next review are different events. The comparison below defines an actionable-condition probability; the numerical illustration uses declared hypothetical inputs. The related door-and-elevator study develops a particular Bayesian proxy-flow model, age/cycle service-condition thresholds, and maintenance scheduling [20]. The present paper shares

Table 3. Assumption register for the hypothetical station. None of these relationships is estimated or validated in the present paper.

Inferential link	Assumption required	Evidence that could challenge it
Station flow to device use	Coverage, routing, and device-choice assumptions apply during the interval represented by the observations.	Local passage observations, closures, or changed platform use.
Device use to exposure	The selected mechanism and operating mode explain the usage quantity being accumulated.	Cycle/run-time records, controller settings, or changed operating practice.
Exposure to condition	Relevant initial-state, degradation, and maintenance-effect assumptions are credible for this asset.	Inspection findings, component history, or signs inconsistent with the assumed mechanism.
Condition to future event	A specified event, horizon, and applicable conditional model are available.	Dated event/inspection records and checks of predictive calibration.
Beliefs to action	Condition probability, detection, net response value, inspection burden, and constraints justify the comparison.	Inspection findings, response outcomes, operator review, and access dependencies.

its station setting and proxy motivation, while developing an applicability checklist and evidence assessment procedure. Its two-door inspection comparison is hypothetical; no technical equations, numerical results, or figures are reproduced from the companion. A service-condition threshold in that work should not be read as a calibrated physical-failure probability here.

Figure 2 summarises the conditional structure. Its arrows identify interpretive dependencies; they do not establish that the available inputs identify every downstream quantity.

5.3. A symbolic two-door inspection comparison

Consider doors A and B on alternative concourse-to-platform routes. The reference policy π_0 retains the existing admissible schedule. Policy π_A uses the optional slot to inspect A early and performs the predefined service response after a positive finding; after a negative finding it retains the schedule. Policy π_B does the same for B. All policies have the same available capacity and service obligations, although π_0 leaves the optional slot unused. No mandatory requirement determines the order in this example. If it does, that requirement takes precedence over the following comparison.

Here an actionable condition is one that a reference inspection could confirm and for which the predefined service response is applicable. For door $i \in \{A, B\}$ and a plausible model m, let $q_i(m)$ be the probability that this condition is present at early inspection, and $d_i(m)$ the probability that the actual inspection detects it when present. Let $k_i(m) \geq 0$ be the incremental inspection burden, incurred regardless of the finding. Let $v_i(m)$ be the conditional expected net loss avoided by the service response after a true positive, relative to π_0. It includes response effectiveness, service effort, and disruption within the common window, excluding the inspection burden already counted in k_i. The losses use one consistent scale; no industrial cost or failure-rate estimate is supplied.

Assume for this bounded comparison that false positive results do not occur. Inspection without a response is assumed to leave the physical state and schedule unchanged. A true positive has probability $q_i d_i$ and conditional net benefit v_i; the remaining outcomes retain the reference policy. Conditioning on these outcomes in Equation 2 gives

$$G_i(m) = q_i(m)d_i(m)v_i(m) - k_i(m), \qquad R_m(\pi_i) = R_m(\pi_0) - G_i(m). \tag{3}$$

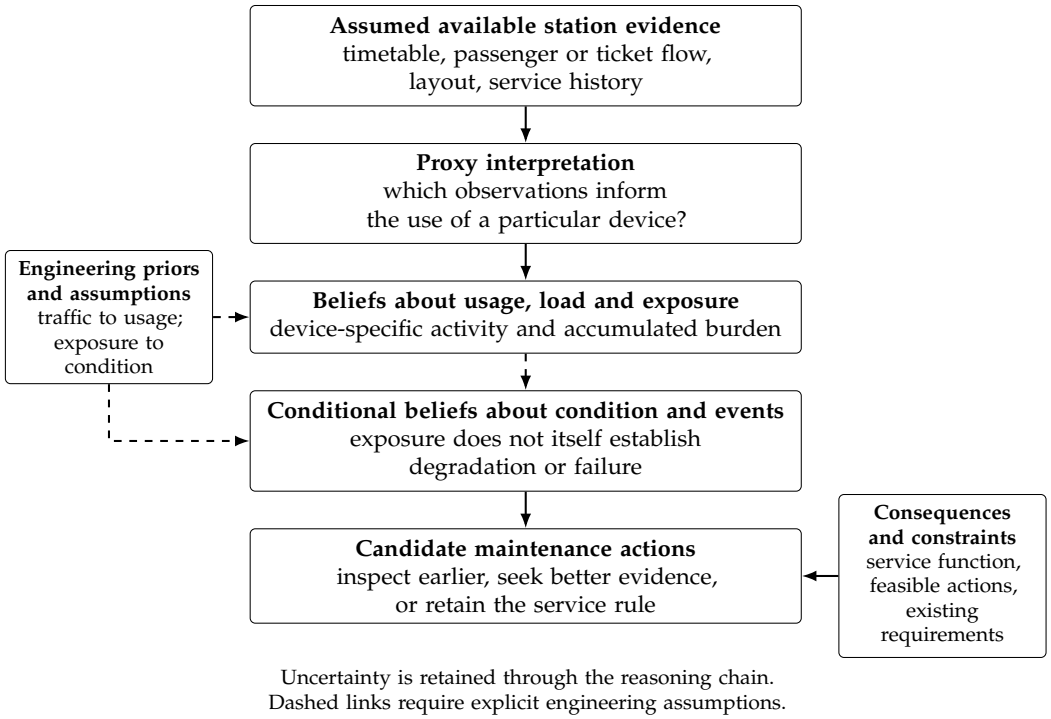


Figure 2. Schematic evidence-to-decision structure for the hypothetical station. Solid arrows trace the reasoning sequence and consequence inputs; dashed arrows highlight the engineering-assumption links shown explicitly. Assumptions and uncertainty also apply to the solid links. The diagram contains no measured or estimated station outputs.

Thus G_i is the expected improvement from a complete inspection-and-response policy. If false positives are possible, subtract $(1 - q_i) f_i h_i$, where f_i is the probability of a positive result conditional on the actionable condition being absent under E, m, and h_i the conditional extra loss of the erroneous response. The compact form also requires either negligible effects on the other door or inclusion of those effects in the conditional loss differences. It does not require independent door states. Negative findings leave any undetected condition in the baseline consequences; they do not certify a healthy door.

Suppose the routing, grouping, and controller assumptions support greater historical cycle exposure for A. Under comparable histories and an assumed increasing exposure–condition relationship, this may support $q_A > q_B$. That additional relationship is not learned from the traffic proxy. For uncertain exposure, ordering mean exposure alone need not order q_i under a nonlinear relationship; the condition probability must account for the exposure distribution. With equal inspection burdens and positive $d_i v_i$, Equation 3 yields the explicit reversal condition

$$G_B > G_A \iff \frac{d_B v_B}{d_A v_A} > \frac{q_A}{q_B}, \qquad q_A > q_B > 0. \tag{4}$$

Equal positive detection probabilities and response values preserve A's order under the equal-burden assumption. A lower-exposure door can therefore warrant earlier inspection when effective detection or the benefit of the available response is sufficiently greater. Either early policy must also improve on retaining the schedule: ranking two negative gains does not justify either inspection.

For a declared nonempty finite set $\mathcal{M}$ of plausible models, a strict preference for A over both B and the reference schedule holds throughout the set exactly when

$$\min_{m \in \mathcal{M}} \{G_A(m) - G_B(m)\} > 0, \qquad \min_{m \in \mathcal{M}} G_A(m) > 0. \tag{5}$$

Interchanging A and B gives the corresponding condition for B. The same statement applies to a compact continuous parameter set when the gains are continuous. A sign change defeats a common order; a tie defeats a strict preference but may permit weak dominance. The inequalities define sensitivity regions; Section 5.4 evaluates them with explicitly hypothetical inputs.

Now suppose dated route records show that a platform reassignment was already in force during the assessed exposure period. The original allocation assigned A passengers who used an adjacent open passage. Correcting that historical interpretation revises the evidence for q_A; a later rerouting could not erase accumulated cycles. If the revised plausible models no longer satisfy Equation 5 for either door, the planner withdraws the differentiated recommendation and retains π_0. B does not acquire priority merely because A's rationale failed. The action record identifies local passage observations and dated route confirmation as possible reasons to reconsider before the next review.

This fallback is the operational rule adopted for the example, not a theorem that π_0 minimizes worst-case loss. It applies only while the existing schedule remains admissible. If an early visit must be allocated, its documented default order or tie-break replaces the unused-slot option. A counting exercise is requested only if it can resolve a relevant comparison in time; no information-value number is asserted without its possible observations, likelihoods, burden, and resulting actions.

The example also clarifies the treatment of maintenance records. It conditions on assumed current condition probabilities and does not estimate them from censored lifetimes. In an empirical application, earlier replacement could censor a lifetime, selective inspection could change which conditions become recorded, and service could change the state itself. Those processes would belong in estimation of q_i and the response consequences. Treating missing or selectively collected records as healthy observations would not justify the inputs to Equation 3.

5.4. Numerical illustration and continuous decision boundaries

Write dimensionless accumulated exposure as $x_i = X_i / X_{\mathrm{ref}}$, where X_i is historical cycle exposure and $X_{\mathrm{ref}} > 0$ is a common reference exposure. Two interpretations of the routing records assign $(x_A, x_B) = (1, 0.5)$ under m_1 and $(0.35, 0.5)$ under m_2. The latter represents passengers diverted from A to a third open passage during the assessed period. It does not transfer them automatically to B. The illustrative condition model is $q_i = 0.1 + \beta x_i$, with $\beta \in [0, 0.5]$. Inspection sensitivity is $d_B = 0.9$ and $d_A \in [0.25, 0.9]$; the common net response value is $v_i = 10$ and inspection burden $k_i = 1$, in arbitrary loss units. False positives are excluded as in the compact model. These values are declared assumptions, not station measurements, elicited uncertainty bounds, or estimated inspection performance. Every probability remains within $[0, 1]$ over the stated ranges.

Substitution into Equation 3 gives

$$\begin{aligned} G_A(m_1) &= d_A(1 + 10\beta) - 1, \quad G_A(m_2) = d_A(1 + 3.5\beta) - 1, \\ G_B &= 4.5\beta - 0.1, \qquad\qquad c(\beta) = \frac{0.9(1 + 5\beta)}{1 + 10\beta}. \end{aligned} \tag{6}$$

Here $c(\beta)$ is the A/B tie boundary under m_1: A has greater gain above it and B below it. B improves on the baseline exactly when $\beta > 1/45$. A's baseline boundary under m_1 is $d_A = 1/(1 + 10\beta)$. Both comparisons matter near $\beta = 0$, where a higher-ranked early inspection can still have negative gain.

At a fixed pair (β, d_A), model m_2 gives B greater gain than A whenever $\beta > 0$. Consequently, across the set $\{m_1, m_2\}$, a strict preference for B over both alternatives holds exactly when $\beta > 1/45$ and $d_A < c(\beta)$. No point in the declared rectangle gives a strict

Table 4. Exact hypothetical inspection comparisons. Gains use arbitrary loss units. The m_1/m_2 column lists the loss-minimizing policies within each model; 0 denotes the existing schedule and braces denote ties. The final column applies the strict common-preference rule to both routing models. A dagger marks its stipulated fallback, which need not minimize loss.

β	d_A	$G_A(m_1)$	$G_A(m_2)$	G_B	m_1/m_2	Both models
0	0.9	-0.1	-0.1	-0.1	$0/0$	0
0.2	0.9	1.7	0.53	0.8	A/B	$0^\dagger$
0.2	0.4	0.2	-0.32	0.8	B/B	B
0.2	0.6	0.8	0.02	0.8	$\{A,B\}/B$	$0^\dagger$
$\frac{1}{45}$	0.9	0.1	-0.03	0	$A/\{0,B\}$	$0^\dagger$
0.5	0.9	4.4	1.475	2.15	A/B	$0^\dagger$

preference for A under both interpretations. At $d_A = c(\beta)$ with $\beta > 1/45$, B is weakly preferred throughout the set but ties A under m_1; the strict rule does not declare a unique common preference. The baseline weakly dominates both early policies throughout the routing set exactly when $\beta \leq 1/45$ and $d_A \leq 1/(1+10\beta)$.

Table 4 illustrates the boundaries with exact calculations. At $\beta = 0.2$ and equal detection $d_A = d_B = 0.9$, the original routing interpretation gives gains 1.7 for A and 0.8 for B, agreeing with its exposure order. Reducing d_A to 0.4 gives gains 0.2 and 0.8, reversing that order while keeping both inspections beneficial under m_1. At $d_A = 0.6$, the gains tie under m_1. The corrected route m_2 favors B in all three cases. If dated evidence establishes m_2 and excludes m_1, B is supported. Keeping both routes plausible leaves disagreement at equal detection, but both favor B when $d_A = 0.4$.

The equal-detection row also exposes a limitation of the adopted fallback. B has gain 0.8 under both routing interpretations, even though A's gain changes from 1.7 to 0.53. Retaining π_0 because neither early policy is strictly best throughout the set therefore forgoes a positive gain available under both models. A lack of unanimous preference is not a lack of beneficial action. The procedure's conservative fallback is distinguishable from a rule that maximizes a guaranteed gain or trades off the model-specific losses. An operator adopting it should acknowledge that cost; operational deployment requires an explicit rule for resolving such conflicts.

Robustness across routing interpretations at fixed parameters must also be distinguished from robustness across the entire declared parameter rectangle. The gains and pairwise differences are bilinear or affine, so their extrema on each rectangle occur at corners. Across both routes and the full parameter ranges, A's gain ranges from -0.75 to 4.4, while B's ranges from -0.1 to 2.15. Neither early policy is beneficial everywhere, and the baseline is not best everywhere either. The calculation therefore supports conditional decision regions, rather than a single preferred policy over all stated assumptions. The table and exact rational results are generated by a deterministic Python calculation from a single input file; no random simulation or fitted data are used.

5.5. Population interpretation and scope

At station level, the planner can group tasks by access window or functional area while preserving these limits. Pooling usage evidence requires comparable mechanisms and a justified sharing structure; combining doors, elevators, and escalators merely because they share a zone does not supply one. Aggregated exposure may inform a workload discussion while leaving individual inspection order unresolved. Shared routing errors, reporting practices, and intervention histories can affect an entire group, so increasing the number of assets need not resolve the dominant uncertainty.

A spatial summary can support this discussion without becoming a map of physical damage. For example, the planner could distinguish a group whose exposure is plausibly concentrated on a busy access route from a group with uncertain allocation among alternative routes. This may help organize a local inspection visit or identify where a short counting exercise would clarify the evidence. The summary should retain both the assumed route structure and the origin of its uncertainty. A shaded zone labelled as a hotspot would otherwise invite a stronger condition interpretation than the underlying information supports.

At population scale, the same assessment must identify shared evidence, admissible coordinated actions, and the loss being compared. Equation 1 explains why a shared source can leave total burden uncertain even when a pairwise contrast is stable. A coordinated policy may also change route use, requiring a joint consequence assessment beyond the separate inspection policies considered here.

5.6. Communicating the basis of a maintenance choice

The planner-facing output should distinguish a supported exposure contrast, a comparison sensitive to routing assumptions, and a comparison that the evidence cannot resolve. These are different reasons for attention even when presented in the same work list. An inspection request should state what is suspected, which evidence supports it, and what finding could change the next action. A request for better routing information should identify the disputed link rather than assigning the asset an unexplained high-uncertainty score. Communicating uncertainty in this form makes the model contestable by service personnel who can recognize a changed controller setting, an omitted intervention, or an inaccessible route. The assumption table and numerical policy comparison illustrate the content of such an output rather than prescribing a software interface.

6. Discussion

6.1. Relationship to established approaches

Small-data PHM already addresses economic and operational constraints, uncertainty, maintenance records, fleet information, and physical knowledge [4]. Usage-based maintenance, soft sensing, and probabilistic inspection planning also provide methods directly relevant to the station problem. Table 5 compares representative formulations through their questions and evidence requirements. Its final column identifies where these established methods enter the proposed assessment; it does not assign exclusive capabilities to LD-PdM.

The specific output here is a sequence of checks that a reader can apply before accepting a maintenance recommendation: classify the information regime, locate the assumptions between evidence and condition, and compare complete actions under competing interpretations. The two-door calculation exposes the detection and response-value assumptions needed to turn an exposure order into an inspection preference. Its expected-loss algebra and sign comparison are applications of established decision analysis. The contribution rests on whether the checklist and procedure make this assessment usable, rather than on methodological priority for those ingredients.

These perspectives can coexist in one application. A small condition dataset may support one decision adequately while leaving another dependent on indirect evidence. A soft sensor or physical usage model can supply an input to the proposed procedure, and a POMDP can implement a sequential policy when its model is justified. The checklist does not exempt any method from validating the relation between its observations and target.

Digital representations can retain asset identity, topology, histories, and assumptions [21–23]. Their role here is to preserve the provenance of a recommendation and the record

Table 5. Representative neighboring approaches and their role in the proposed assessment. The comparison concerns organizing questions and assumptions, not mutually exclusive method classes.

Approach	Question and required evidence	Role in the procedure
Small-data PHM [4]	How can PHM use limited quantity or quality of evidence, including records and physical knowledge?	Distinguish sample limitations from weak condition information before choosing learning methods.
Usage/load-based maintenance [8]	How do usage and loads affect condition and maintenance intervals under a physical model?	Check the usage-to-condition assumptions, including cases where usage itself is inferred.
Soft sensing [9]	Can other measurements and a validated relationship estimate an unavailable process quantity?	Identify the target of each proxy and the evidence for applying its relationship.
POMDP and information-value maintenance [5,6]	Which observation and maintenance policy is justified by condition uncertainty, costs, and consequences?	Compare complete inspection policies after specifying actions, observation quality, and constraints.

that would trigger its revision. A full digital twin is not a prerequisite for applying the checklist or recording the finite policy comparison.

6.2. What a later evaluation should establish

The station example derives conditional policy comparisons from stated assumptions. The gain and reversal inequalities can be checked algebraically, but their practical value depends on whether the assumed condition, detection, and response relationships describe the asset. A declared synthetic model and comparison design can establish conditional policy differences. Assessing predictive calibration or realized industrial savings in the target setting additionally requires empirical observations.

The following are recommendations for evaluating an implementation, rather than a committed experimental programme. Evaluation should separate the quality of proxy interpretation from the quality of the resulting choice. Where a quantity is observed for validation, useful checks include whether uncertainty estimates are calibrated and whether proxy relationships transfer across operating contexts. At the decision level, relevant outcomes include consequence-weighted missed events, unnecessary inspection or intervention, and the value of acquired information under equal resources. Ranking stability and predictive accuracy are useful supporting measures, but neither alone establishes decision benefit.

Exact reconstruction of condition is unnecessary when all plausible states support the same admissible action. Close to the reversal boundary in Equation 4, however, a small change in detection or response value can change the preferred door. Evaluation should therefore test the uncertain distinctions that can change the choice, alongside the accuracy of the estimated quantities.

The present exact calculation isolates routing interpretation and inspection sensitivity under a fixed hypothetical condition model. A later stochastic experiment could examine noisy counts and shared proxy bias. Its conclusions would remain conditional on the generated environment, and an exposure-based inspection objective would not establish failure reduction. Empirical work would additionally need observation and intervention records adequate to distinguish changing condition from changing recording practice. A calibrated failure model or industrial sensing break-even estimate therefore remains a separate research task.

6.3. Scope and limitations

The analysis provides an assessment procedure, conditional analytical results, and an exact numerical illustration with hypothetical inputs. It supplies no fitted exposure–condition relationship, calibrated detection model, station cost estimate, or empirical maintenance comparison. The two-door calculation assumes a feasible service response, a common loss scale, and an admissible reference schedule. Its compact form excludes false positives and either excludes cross-door intervention effects or includes them in the conditional consequences. The stated extension shows where erroneous-response losses enter. Conclusions about a physical-failure policy would require a separate event and transition model.

The available evidence may leave several models compatible with the same records. In the example, passenger records alone supply no observations of the targeted condition or inspection sensitivity, so they cannot estimate q_i and d_i without additional relationships or outcomes. Even observations of the positive-finding rate alone identify only the product $q_i d_i$ under the zero-false-positive simplification; separating prevalence from detection requires additional information. The procedure therefore retains competing assumptions and bounds its preference to the declared model set. An omitted interpretation can invalidate a preference that was stable within that set.

Application also requires reconsidering the acquisition constraint. Better condition information may be worthwhile when it changes an important action enough to justify its lifecycle burden. Rerouting, refurbishment, altered logging, or selective inspection can change which interpretation of existing records is credible. An operational implementation needs a review trigger and a record of such changes. The conceptual classification itself neither establishes that a new monitoring system is economical nor justifies relaxing safety, accessibility, or service obligations.

7. Conclusions

Limited-Data Predictive Maintenance characterizes consequential maintenance choices for which feasible condition information remains weak and indirect evidence materially informs the assessment of future needs. Its applicability depends on the choice, horizon, and evidence, so neither a small dataset nor a particular asset class determines membership. The title's asymmetry is the mismatch between the consequences of a choice and the information available to support it.

The checklist, five-step procedure, and station example provide a concrete way to examine such a proposal. The analytical and numerical comparisons show that an exposure-based order requires assumptions about actionable condition, detection, response value, and inspection burden. In the stated example, lower detection at the more exposed door reverses the inspection order. Routing alternatives can also prevent a common strict preference while leaving B beneficial under both models. The fallback to the existing schedule therefore has an explicit opportunity cost: an operator must distinguish inability to name one uniformly best inspection from inability to improve on the baseline.

The analysis establishes these conditional implications without estimating station performance. Future evaluation should test the proxy relationships against independent observations and assess the consequences of complete inspection policies under equal resource constraints. Such work would determine whether the proposed assessment improves maintenance decisions in a particular operating setting.

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Abbreviations

PdM	Predictive maintenance
LD-PdM	Limited-Data Predictive Maintenance
PHM	Prognostics and health management
RUL	Remaining useful life
HVAC	Heating, ventilation, and air-conditioning
POMDP	Partially observable Markov decision process

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